

Plantee un sistema de ecc. recurrente que le permita calcular los sucesivos nros de Bernoulli ($n=1$)

$$\hat{B}_V = B_V(0) \rightarrow \text{coeficiente independiente de los polinomios de Bernoulli.}$$

Si $\lambda=0$ y $x=0$ (condición para nro de Bernoulli)

$$b(1; \tau, 0) = \frac{\tau e^{\tau \cdot 0}}{(e^{\tau} - 1)^1} = \frac{\tau}{e^{\tau} - 1} = \sum_{v=0}^{\infty} \frac{\tau^v}{v!} B_V(0)$$

$$\tau = (e^{\tau} - 1) \sum_{v=0}^{\infty} \frac{\tau^v}{v!} B_V(0)$$

$$\tau = \sum_{r=1}^{\infty} \frac{\tau^r}{r!} \cdot \sum_{v=0}^{\infty} \frac{\tau^v}{v!} B_V$$

$$\tau = \left(\frac{\tau}{1!} + \frac{\tau^2}{2!} + \frac{\tau^3}{3!} + \frac{\tau^4}{4!} + \dots \right) \left(\frac{B_0}{0!} + \tau \frac{B_1}{1!} + \frac{\tau^2}{2!} B_2 + \frac{\tau^3}{3!} B_3 + \frac{\tau^4}{4!} B_4 + \dots \right)$$

$$\tau = \tau B_0 + \tau^2 \left(\frac{B_0}{2! 0!} + \frac{B_1}{1! 1!} \right) + \tau^3 \left(\frac{B_0}{3! 0!} + \frac{B_1}{2! 1!} + \frac{B_2}{1! 2!} \right) +$$

$$+ \tau^4 \left(\frac{B_0}{4! 0!} + \frac{B_1}{3! 1!} + \frac{B_2}{2! 2!} + \frac{B_3}{1! 3!} \right)$$

Notar patrón $\tau^k \left(\frac{B_0}{k! 0!} + \frac{B_1}{(k-1)! 1!} + \dots + \frac{B_{k-1}}{1! (k-1)!} \right)$

Revelar igualdad

1) $\tau = \tau \cdot B_0 \rightarrow B_0 = 1$

2) $\frac{B_0}{2! 1!} + \frac{B_1}{1! 1!} = 0 \rightarrow \frac{1}{2} + B_1 = 0 \rightarrow B_1 = -\frac{1}{2}$

$$3) \frac{B_0}{3!} + \frac{B_1}{2!1!} + \frac{B_2}{1!2!} = 0$$

$$\frac{1}{6} - \frac{1}{4} + \frac{B_2}{2} = 0 \rightarrow B_2 = \left(\frac{1}{4} - \frac{1}{6} \right) \cdot 2 = \frac{1}{6} = B_2$$

$$4) \frac{B_0}{4!} + \frac{B_1}{3!1!} + \frac{B_2}{2!2!} + \frac{B_3}{1!3!} = 0$$

$$\frac{1}{24} - \frac{1}{12} + \frac{1}{24} + \frac{B_3}{6} = 0 \rightarrow B_3 = 0$$

Demonstraciones

$$1) B_v^{n+1}(x+1) = B_v^{n+1}(x) + v B_{v-1}^n(x)$$

$$B_v^{n+1}(x+1) - B_v^{n+1}(x) = v B_{v-1}^n(x)$$

$$\Delta_x \left(B_v^{n+1}(x) \right) = v B_{v-1}^n(x) \text{ por propiedad.}$$

$$2) \int_a^x B_v^n(\tau) d\tau = \frac{1}{v+1} \left[B_{v+1}^n(x) - B_{v+1}^n(a) \right]$$

$$\text{NOTAR } D \left(B_{v+1}^n(\tau) \right) = \overline{v+1} B_v^n(\tau)$$

$$\int_a^x \frac{D \left(B_{v+1}^n(\tau) \right)}{v+1} d\tau = \frac{1}{v+1} B_{v+1}^n(\tau) \Big|_a^x$$

$$3) \int_x^{x+1} B_v^1(\tau) d\tau = B_v^{\wedge-1}(x)$$

$$\int_x^{x+1} \frac{D(B_{v+1}^1(\tau))}{v+1} d\tau = B_v^{\wedge-1}(x)$$

$$\frac{1}{v+1} B_{v+1}^1(\tau) \Big|_x^{x+1} = \frac{1}{v+1} \Delta B_{v+1}^1(x)$$

$$\frac{1}{v+1} \cdot \underbrace{v+1}_{\text{propriedade } \Delta} B_v^{\wedge-1}(x) = B_v^{\wedge-1}(x)$$

$$4) \int_x^{x+1} B_v(\tau) d\tau = x^v$$

$$\int_x^{x+1} \frac{D(B_{v+1}^0(\tau))}{v+1} d\tau = \frac{1}{v+1} \Delta B_{v+1}^0(x)$$

$$\frac{1}{v+1} (v+1) B_v^0(x) = x^v$$

$$\text{propriedade } B_v^0(x) = x^v$$

$$\sum_{v=0}^{\infty} \frac{\tau^v}{v!} B_v^0(x) = \frac{\tau^0 e^{\tau x}}{(e^{\tau} - 1)^0} = e^{\tau x}$$

$$\underline{B_0^0(x)} + \tau \underline{B_1^0(x)} + \frac{\tau^2}{2!} \underline{B_2^0(x)} = \underline{1} + \underline{\tau x} + \frac{\tau^2}{2!} \underline{x^2} \dots$$

Ejercicio 3.

$$* \Delta^n B_v^1(x) = v^{(1)} x^{v-n} \quad \text{si } v \geq n$$

$$\begin{aligned} \Delta^{n-1} \Delta B_v^1(x) &= \Delta^{n-1} \left(v \cdot B_{v-1}^1(x) \right) = v^{(2)} \Delta^{n-2} \left(B_{v-2}^{1-2}(x) \right) \dots \\ &= v^{(n)} \cdot \Delta^0 \left(B_{v-n}^0(x) \right) = v^{(n)} x^{v-n} \end{aligned}$$

$$* \Delta^n B_v^1(x) = 0 \quad \text{si } v < n$$

$$\begin{aligned} \Delta^n B_v^1(x) &= \Delta^{n-v} \left(\Delta^v B_v^1(x) \right) = \Delta^{n-v} \left(v^{(v)} B_0^{1-v}(x) \right) \\ &= v^{(v)} \underbrace{\Delta^{n-v} \left(\underbrace{B_0^{1-v}(x)}_{=1} \right)}_{=0} = 0 \end{aligned}$$

$$* \Delta^c B_c^c = 0 \quad \text{since } v > n$$

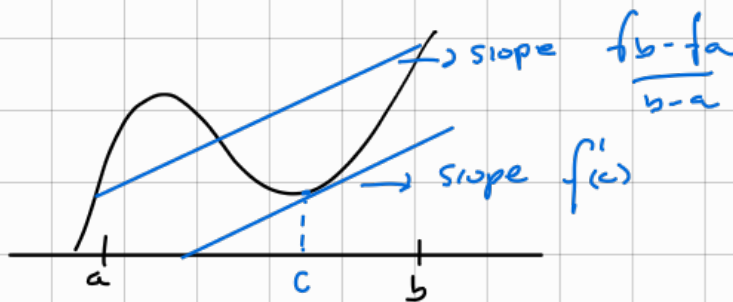
$$* \Delta^7 B_9^7 = q^{(7)} \cdot B_2^0(0) = q^{(7)} 0^2 = 0$$

Demostrar que todos los p. de Bernoulli de orden 1 y grado mayor o igual a 1 tienen al menos una raíz en el intervalo $[0,1]$

Teorema del Valor Medio:

Si f es continua en $[a,b]$ y diferenciable en (a,b) entonces existe un punto c en (a,b) tal que la recta tangente en el punto c es paralela a la recta secante que pasa por $(a; f(a))$; $(b; f(b))$

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$



$$f'(c)(b-a) = f(b) - f(a)$$

$$f'(c) \cdot (b-a) = \int_a^b f'(x) dx$$

Luego: $\int_0^1 B_\nu(x) dx = 0$ si $\nu \geq 1$

$$B_\nu(c)(1-0) = 0 \Rightarrow B_\nu(c) = 0 \text{ con } c \in [0,1]$$

7) Expresar $G(x) = 3x^{(3)} + 4(x-1)^{(2)} - 5x - \frac{1}{2}$ en términos de P.B.

Note: $B_n^{(n+1)}(x+1) = x^{(n)}$

proof: $B_{\nu}^{(n+1)}(x) = \left(1 - \frac{\nu}{n}\right) B_{\nu}^{(n)}(x) + \nu \left(\frac{x}{n} - 1\right) B_{\nu-1}^{(n)}(x)$

$$B_n^{(n+1)}(x) = \left(1 - \frac{n}{n}\right) B_n^{(n)}(x) + n \left(\frac{x}{n} - 1\right) B_{n-1}^{(n)}(x)$$

$$B_n^{(n+1)}(x) = 0 + (x-1) B_{n-1}^{(n)}(x)$$

$$B_n^{(n+1)}(x) = (x-n) \cdot B_{n-1}^{(n)}(x)$$

$$B_n^{(n+1)}(x) = (x-n) \cdot (x-n+1) B_{n-2}^{(n)}(x)$$

$$B_n^{(n+1)}(x) = (x-n)(x-n+1) \dots (x-1) \underbrace{B_0^{(n)}(x)}_{B_0^{(n)}(x)=1}$$

$$B_n^{(1)}(x) = (x-n)(x-n-1) \dots (x-1) = (x-1)^{(n)}$$

$$\text{Luego: } G(x) = 3x^{(3)} + 4(x-1)^{(2)} - 5x - \frac{1}{2}$$

$$G(x) = 3B_3^4(x+1) + 4B_2^3(x) - 5B_1^2(x-1) - \frac{1}{2}B_0^1(x)$$

Ejercicio 8

$$a) \rightarrow D \left[(x-2)^{(3)} + 5x^{(2)} \right] = D \left[B_3^4(x-2) + 5B_2^3(x) \right]$$

$$= 3B_2^4(x-2) + 5 \cdot 2B_1^3(x)$$

$$b) \int (\tau-3)^{(2)} d\tau = \int B_2^3(\tau-3) d\tau$$

$$= \int \frac{D[B_3^3(\tau-3)]}{3} d\tau = \frac{1}{3} \cdot B_3^3(\tau-3) + c$$

Ejercicio 9

$$\text{note } B_0^0(x) = x^0$$

$$a) \sum_{x=0}^7 (x^4 - 5x^2 - 2) = \sum_{x=0}^7 (B_4^0(x) - 5B_2^0(x) - 2)$$

$$= \sum_{x=0}^7 B_4^0(x) - 5 \sum_{x=0}^7 B_2^0(x) - 16$$

$$\text{note } \Delta B_n^1(x) = n B_{n-1}^{1-1}(x)$$

$$= \bar{\Delta}'(B_4^0(x)) \Big|_0^8 - 5 \bar{\Delta}'(B_2^0(x)) \Big|_0^8 - 16$$

$$\text{Si: } \Delta B_5'(x) = 5 \cdot B_4^0(x) \Rightarrow \frac{B_5'(x)}{5} = \bar{\Delta}' B_4^0(x)$$

Luego

$$= \frac{B_5'(x)}{5} \Big|_0^8 - 5 \cdot \frac{B_3'(x)}{3} \Big|_0^8 - 16$$

$B_v'(x)$ se pueden conseguir con recurrencia: $B_1'(x) = x - \frac{1}{2}$

$$b) \sum_{x=1}^n (x+2)^3 = \sum_{x=1}^n \left(x^3 + 3x^2 \cdot 2 + 3x \cdot 2^2 + 2^3 \right)$$

$$= \sum_{x=1}^n x^3 + 6 \sum_{x=1}^n x^2 + 12 \sum_{x=1}^n x + n \cdot 8$$

since $B_v^0(x) = x^v$

$$= \sum_{x=1}^n B_3^0(x) + 6 \sum_{x=1}^n B_2^0(x) + 12 \sum_{x=1}^n B_1^0(x) + 8n$$

since $\Delta B_v^1(x) = v B_{v-1}^0(x) \rightarrow \bar{\Delta}' B_{v-1}^0(x) = \frac{B_v^1(x)}{v}$

$$= \frac{B_4^1(x)}{4} \Big|_1^{n+1} + 6 \frac{B_3^1(x)}{3} \Big|_1^{n+1} + 12 \frac{B_2^1(x)}{2} \Big|_1^{n+1} + 8n$$

Ejercicio 6.

Datos: $B_3^2(x) = x^3 - 3x^2 + \frac{5}{2}x - \frac{1}{2}$

$$B_1^2(x) = x - 1$$

a) Buscar $B_2^2(x)$

$$D(B_3^2(x)) = 3B_2^2(x) = 3x^2 - 3 \cdot 2 \cdot x + \frac{5}{2}$$

Despejando

$$B_2^2(x) = x^2 - 2x + \frac{5}{6}$$

b) $B_3^3(x)$. Propiedad $B_n^{1+n}(x) = (1 - \frac{n}{n}) B_n^1(x) + n(\frac{x}{n} - 1) B_{n-1}^1(x)$

$$B_3^3(x) = \left(1 - \frac{3}{2}\right) \underbrace{B_3^2(x)}_{x^3 - 3x^2 + \frac{5}{2}x - \frac{1}{2}} + 3\left(\frac{x}{2} - 1\right) \underbrace{B_2^2(x)}_{x^2 - 2x + \frac{5}{6}}$$

$$B_3^3(x) = \frac{1}{2} \left(-x^3 + 3x^2 - \cancel{\frac{5}{2}x} + \cancel{\frac{1}{2}} \right) + \frac{3}{2}x^3 - 3x^2 + \cancel{\frac{5}{4}x} - 3x^2 + 6x - \frac{5}{2}$$

$$B_3^3(x) = x^3 - \frac{9}{2}x^2 + 6x - \frac{9}{4}$$

$$D B_3^3(x) = 3 B_2^3(x)$$

$$B_2^3(x) = \frac{1}{3} \left(3x^2 - 9x + 6 \right) = x^2 - 3x + 2$$

$$B_2^3(x) = x^2 - 3x + 2$$

$$\hookrightarrow B_2^3(x) = 2 B_1^3(x)$$

$$B_1^3(x) = \frac{1}{2} \left(2x - 3 \right) = x - \frac{3}{2} = B_1^3(x)$$

$$B_v^{1+v}(x) = \left(1 - \frac{v}{n} \right) B_v^1(x) + v \left(\frac{x}{n} - 1 \right) B_{v-1}^1(x)$$

$$B_3^4(x) = \left(1 - \frac{3}{3} \right) B_3^3(x) + 3 \left(\frac{x}{3} - 1 \right) B_2^3(x)$$

$$B_3^4(x) = (x - 3) B_2^3(x)$$

$$B_3^4(x) = x - 3 \left(x^2 - 3x + 2 \right) = x^3 - 3x^2 + 2x - 3x^2 + 9x - 6$$

$$B_3^4(x) = x^3 - 6x^2 + 11x - 6$$

$$B_2^4(x) = \frac{1}{3} \hookrightarrow (B_3^4(x))$$

$$B_2^4(x) = \frac{1}{3} \left(3x^2 - 12x + 11 \right) = x^2 - 4x + \frac{11}{3}$$

Ejercicio 10

$$\hat{B}_1^{(n)}(x+1) = x^{(n)}$$

$$f(x) = 5(x+4)^{3/2} + \binom{x+1}{3}; \text{ find } \Delta^4 \iint f(x) dx^2 \text{ en } x=2.$$

Paso $f(x)$ a pol de Bernoulli.

$$f(x) = 5 B_2^3(x+5) + \frac{(x+1)!}{3!(x+1-3)!} = 5 B_2^3(x+5) + \frac{(x+1)^{3/2}}{6}$$

$$\text{NOTE } \binom{x+1}{3} = \frac{(x+1)(x)(x-1)}{6} = \frac{(x+1)^{3/2}}{6}$$

$$f(x) = 5 B_2^3(x+5) + \frac{1}{6} B_3^4(x+2)$$

$$\text{Busco } \Delta^4 \iint f(x) dx^2$$

$$\text{Si } D \hat{B}_v^1(x) = v \hat{B}_{v-1}^1(x) \rightarrow \hat{B}_v^1(x) = \underbrace{I(\hat{B}_{v-1}^1(x))}_v$$

$$I(\hat{B}_{v-1}^1(x)) = \frac{\hat{B}_v^1(x)}{v}$$

Luego

$$I^2 f(x) = I^2 \left(5 B_2^3(x+5) + \frac{1}{6} B_3^4(x+2) \right)$$

$$= I \left(\frac{5}{3} B_3^3(x+5) + \frac{1}{24} B_4^4(x+2) \right)$$

$$I^2 f(x) = \frac{5}{12} B_4^3(x+5) + \frac{1}{120} B_5^4(x+2)$$

$$\Delta^4 I^2 f(x) = \frac{5}{12} \left(\Delta \Delta^3 B_4^3(x+5) \right) + \frac{1}{120} \cdot 5 \cdot 4 \cdot 3 \cdot 2 B_{5-4}^{4-4}(x+2)$$

$$\Delta^4 I^2 f(x) = \frac{5}{12} \Delta \left(4 \cdot 3 \cdot 2 \underbrace{B_1^0(x+5)}_{(x+5)^1} \right) + \frac{120}{120} \underbrace{B_1^0(x+2)}_{(x+2)^1}$$

$$= \frac{5}{12} \cdot 24 \cdot (\Delta(x) + \Delta(5)) + (x+2)$$

$$\Delta^4 I^2 f(x) = 10 \cdot (1+0) + x+2 = 12+x$$

$\Delta^4 I^2 f(x) \Big _{x=2} = 12+x \Big _{x=2} = 14$
